# Inferential Statistics:-

Inference; conclusion;

## Statistical Estimation:-

- Population - Entire data
- Sample - Subset of population
- Random sampling: selecting randomly
- Point estimate: generalizing something with single point (SD, Variance).
- Interval Estimate: Giving interval & understanding within the range.
- Bias: Systematic Error
- Variance: Sensitivity to sample.

## Hypothesis Testing:- Hypothesis —

Null Hypothesis (H0)
- Default Assumption
- (what ever I am trying to prove which is right by assuming it but don't know it is right (or) wrong we should prove statistically.

Eg:- New model has equal accuracy.

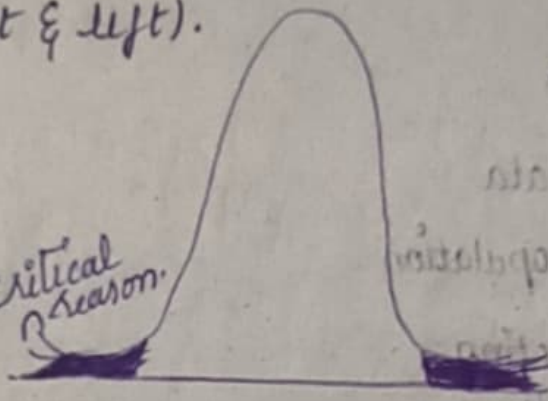
Alternative Hypothesis (H1)
- Testing.
- opposite of Null hypothesis.
- Eg:- New model is more accurate.

### Two-Tailed Test:-

One-Tailed Test:- Critical region on single side. (either right or left)

Two-Tailed Test:- Critical region on both sides. (right & left).

=>




Redcolour - Critical region: either right or left

## Testing Hypothesis & Workflow:-

1. Define $H_0$ and $H_1$.
   ↳ state null & alternative hypothesis clearly.
2. Select significance level ($\alpha$) (Threshold level)
   ↳ set threshold (commonly 0.05) for rejecting $H_0$.
3. choose Test statistic:- z-test, t-test, chi-squared based on data
   ↳ chi-squared test (categorical data)
   ↳ z-test } numerical
   ↳ t-test } data.
   ↳ sample size is < 30 (low) ⇒ (t-test)
   ↳ sample size is > 30 (high) ⇒ (z-test)
   ↳ If we know the population variance (z-test)
   ⇒ If we don't know " " (t-test)
4. compute P value ($P < \alpha$ (reject $H_0$ & choose $H_1$ (alternate)
   ~~failed to reject (accept it)~~
5. Rule Decision - Reject $H_0$ if $P \leq \alpha$; otherwise fail to reject.

| Aspect | Confidence Interval | Hypothesis Test. |
| --- | --- | --- |
| Output | Range of possible values. | Binary decision (T/F) |
| Question | "what is the value"? | "Is there any effect" |
| Key Parameter. | confidence level $(1-\alpha)$ (95%) (100-5) | significance level $(\alpha)$ (5%) |

$(1-\alpha)$ & $(\alpha)$ both are interdependent.

| | | |
| --- | --- | --- |
| ML Use case. | Estimating model metrics. | A/B testing, feature tests. |
| Connection | CI excludes Ho → rejects Ho. | $P \leq \alpha \rightarrow$ reject Ho |

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Compute p value :-

- If $p < \alpha \Rightarrow$ reject $H_0$ & choose $H_1$ (alternative hypothesis)
  If $p \geq \alpha \Rightarrow$ reject $H_1$ & choose $H_0$ (null hypothesis)

Type1 error :- If null hypothesis is true but unfortunately we are rejecting it called Type 1 error.

Type2 Error: If null hypothesis is ~~false~~ but we are accepting it called type 1 Error

Assumption:-

| | | Yes | no |
|---|---|---|---|
| Actual | yes | true | mistake |
| | no | mistake | true |

Decision:

| | | Accept null hyp | Reject null hyp |
|---|---|---|---|
| Actual | Null Hyp True | True | Type1 Error |
| | Null Hyp False | Type2 Error | True |

⇒ Confidence level :- Range estimated an unknown population parameter with a stated confidence level $(1-\alpha)$.

when to use which :-

- Use confidence intervals when you need to estimate magnitude & precision.
- Use hypothesis tests when you need to clear yes/no decision about an effect.

Provide both direction & magnitude. the complete for data-driven decisions.